

Statistical Inference in Relative Risk and Odds ratio

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Background & Motivation

In fields such as public health, epidemiology, biostatistics, and genetics, a central question researchers face is whether a particular exposure or treatment influences a given outcome. For instance, does a family history of breast cancer increase an individual's likelihood of developing the disease? Or does a therapeutic drug meaningfully improve patient recovery? Answering these questions rigorously requires a principled statistical framework for quantifying the relationship between exposure and outcome.

Relative Risk and Odds Ratio

Two of the most widely used measures for this purpose are the **relative risk (RR)** and the **odds ratio (OR)**. Both metrics compare the probability of an outcome across two groups: those exposed to a factor and those who are not. Let $\pi_E = P(\text{Disease} \mid \text{Exposure})$ and $\pi_{NE} = P(\text{Disease} \mid \text{No Exposure})$. The relative risk is defined as the ratio of the probability of the outcome given exposure to the probability of the outcome given no exposure:

$$RR = \frac{\pi_E}{\pi_{NE}}$$

An RR greater than 1 suggests that exposure is associated with a higher risk, while an RR less than 1 suggests a protective effect.

The odds ratio, by contrast, compares the odds of the outcome in each group:

$$OR = \frac{\frac{\pi_E}{1-\pi_E}}{\frac{\pi_{NE}}{1-\pi_{NE}}}$$

While RR has a more direct and intuitive interpretation, OR is particularly valuable in case-control study designs where outcome prevalence cannot be directly estimated from the sample.

In practice, both measures are estimated empirically from a **2x2 contingency table**, which cross-tabulates exposure status against outcome status across a sample population.

Table 1: 2x2 Table: Exposure VS Disease Outcome

	Disease	No Disease
Exposed	a	b
Not Exposed	c	d

Letting a,b,c,d denote the cell counts as laid out above, the sample estimates become $\pi_E = \frac{a}{a+b}$ and $\pi_{NE} = \frac{c}{c+d}$, yielding $\hat{RR} = \frac{\pi_E}{\pi_{NE}}$ and $\hat{OR} = \frac{ad}{bc}$.

Project Purpose

While RR and OR provide useful point estimates of association, a single estimate alone is insufficient for rigorous statistical inference. We also need to quantify the uncertainty around it. This is where confidence intervals (CIs) become essential. A confidence interval provides a range of plausible values for the true population parameter, constructed such that if we were to repeat the study many times, the interval would contain the true value at a specified confidence level. In this project, we focus specifically on 95% confidence intervals, meaning that 95% of such constructed intervals would contain the true RR or OR across repeated samples. Understanding how accurately and precisely different methods construct these intervals is therefore central to the goals of this project.

The core methodological question driving this project is: when computing confidence intervals for non-linear statistics like RR and OR, does it matter which construction method we use? To answer this, this directed reading project compares two approaches, the **delta method** and **bootstrapping**, through a simulation study conducted in R. Each method has different theoretical assumptions and computational demands, and our goal is to evaluate them on two criteria: *coverage accuracy* (does the interval contain the true parameter at the nominal 95% rate?) and *precision* (how wide is the interval?).

The simulation is grounded in real data drawn from Manders et al. (2017), Family history and risk of breast cancer: an analysis accounting for family structure (PubMed). Specifically, we extract the 2×2 table from Table 3 of the paper, which records breast cancer diagnoses stratified by family history:

Table 2: Observed Breast Cancer Diagnoses by Family History
(Manders et al., 2017)

	Diagnosed	Not Diagnosed
Family History: Yes	432	1301
Family History: No	87787	14218

From this table, we derive the observed conditional probabilities $\hat{\pi}_E = \frac{432}{432+87787}$ and $\hat{\pi}_{NE} = \frac{1301}{1301+14218}$, yielding a ground truth relative risk $RR_{\text{simulation}} = \frac{\hat{\pi}_E}{\hat{\pi}_{NE}} = 17.12$, which serves as the assumed population parameter throughout the simulation.

Methodology

Because the Manders et al. (2017) study is a cohort study where disease probabilities can be estimated directly from the data, we focus exclusively on RR rather than OR.

Delta Method

The delta method is an analytical technique for approximating the variance of a non-linear function of a random variable using a first-order Taylor expansion. Since RR is a non-linear function of the underlying proportions $\hat{\pi}_E$ and $\hat{\pi}_{NE}$, its variance cannot be derived directly.

We begin by noting that since $\hat{\pi}_E$ and $\hat{\pi}_{NE}$ are sample proportions drawn from binomial distributions, their variances follow from standard binomial theory:

$$\text{Var}(\hat{\pi}_E) = \frac{\pi_E(1 - \pi_E)}{|E|}, \text{Var}(\hat{\pi}_{NE}) = \frac{\pi_{NE}(1 - \pi_{NE})}{|NE|}$$

where $|E|$ and $|NE|$ denote the number of exposed and unexposed individuals, respectively. By the Central Limit Theorem, the joint distribution of (π_E, π_{NE}) is approximately bivariate normal with covariance matrix:

$$\Sigma = \begin{bmatrix} Var(\hat{\pi}_E) & 0 \\ 0 & Var(\hat{\pi}_{NE}) \end{bmatrix}$$

The off-diagonal entries are zero because the two groups are independent. Now, since $RR = g(\pi_E, \pi_{NE}) = \frac{\pi_E}{\pi_{NE}}$, we apply the delta method, which states that for a smooth function g :

$$g(\hat{\theta}_n) \approx N(g(\theta), \nabla g(\theta)^T \Sigma g(\theta))$$

The gradient of $g(x, y) = \frac{x}{y}$ is $\nabla g = [\frac{1}{y}, -\frac{x}{y^2}]$, which evaluates at (π_E, π_{NE}) gives $\nabla g = [\frac{1}{\pi_{NE}}, -\frac{\pi_E}{\pi_{NE}^2}]$. Applying the delta method formula yields the following variance approximation for $\hat{RR}$:

$$\psi^2 = \frac{\pi_E(1 - \pi_E)}{\pi_{NE}^2 \cdot |E|} + \frac{\pi_E^2(1 - \pi_{NE})}{\pi_{NE}^3 \cdot |NE|}$$

A 95% confidence interval for RR is then constructed as:

$$CI_{RR} = (\hat{RR} - z_{1-\frac{\alpha}{2}} \cdot \hat{\psi}, \hat{RR} + z_{1-\frac{\alpha}{2}} \cdot \hat{\psi})$$

where $z_{1-\frac{\alpha}{2}} = 1.96$ for a 95 confidence interval.

Bootstrapping

Bootstrapping is a resampling-based method that estimates the sampling distribution of a statistic empirically, without relying on distributional assumptions such as normality. Rather than deriving a formula for variance, bootstrapping lets the data speak for itself.

The procedure is straightforward: repeatedly draw a new dataset of the same size as the original with replacement, and recompute $\hat{RR}$ each time. After B resamples, we obtain a collection of bootstrap estimates $\hat{RR}_1^*, \dots, \hat{RR}_B^*$ that approximates the sampling distribution of $\hat{RR}$. The 95% confidence interval is then simply the 2.5th and 97.5th percentiles of this distribution:

$$CI_{BootRR} = (\hat{RR}_{0.025}^*, \hat{RR}_{0.975}^*)$$

In this simulation, we use $B = 2000$ resamples per iteration, because our earlier finding that $B = 100$ produced insufficient coverage of 93.2%, which we attributed to Monte Carlo error from too few resamples.

Simulation design

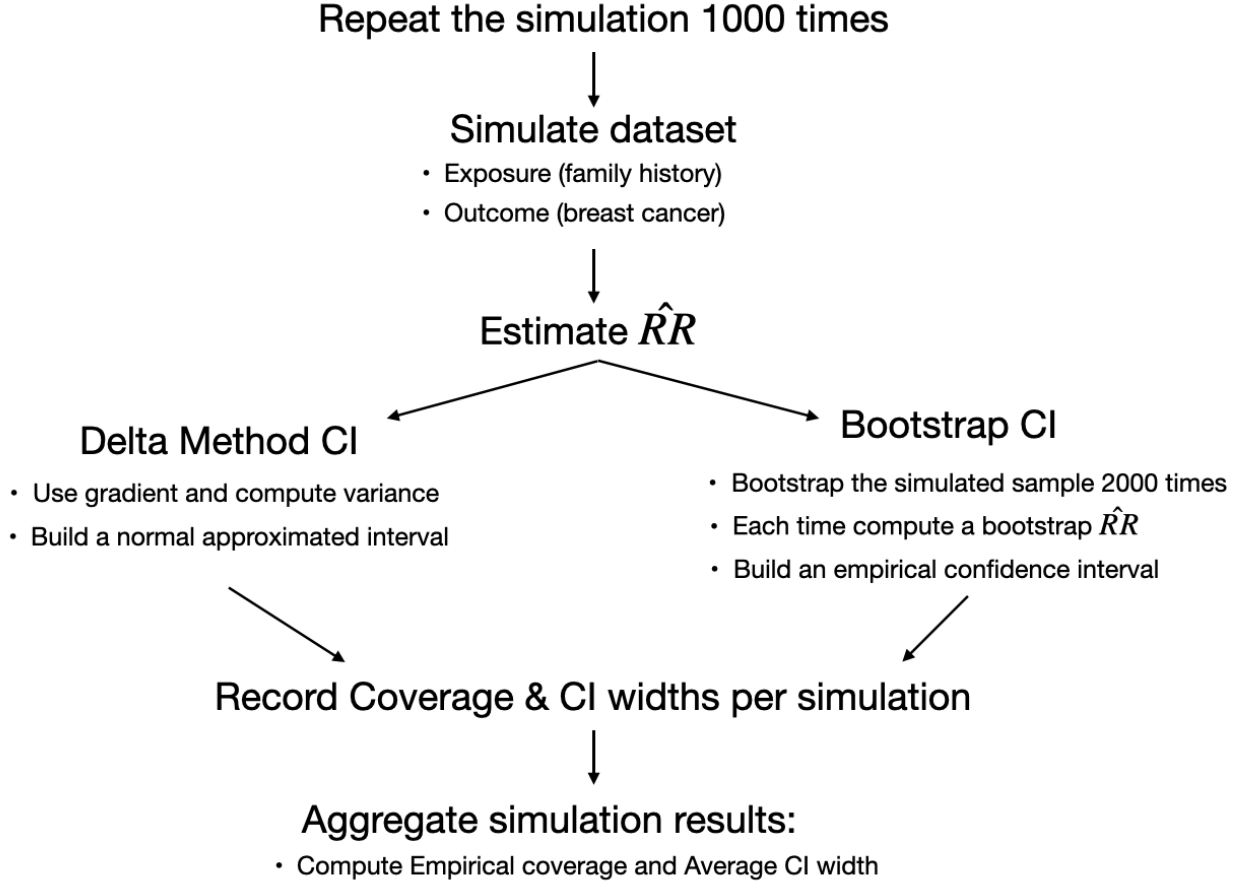


Figure 1: Diagram of Simulation design

To compare the two methods, we conduct a simulation study in R using population parameters derived from the Manders et al. (2017) data, with the ground truth set at $RR_{simulation} = 17.12$. The simulation is repeated 1,000 times. In each replicate, we generate a synthetic dataset of $n = 10,000$ individuals with exposure (family history) and outcome (breast cancer) simulated using π_E and pi_{NE} from the observed data. We then estimate $\hat{RR}$ and construct a 95% CI using both methods, recording the coverage and width of each interval. Final performance is summarized as empirical coverage and mean CI width across all 1,000 replicates, as illustrated in the diagram above.

Results & Comparison

The simulation results are summarized across two performance metrics: coverage and mean CI width. Table 2 below presents the findings for both methods.

In terms of coverage accuracy, the bootstrap method achieves the nominal 95% target exactly, confirming that our earlier undercoverage of 93.2% at $B = 100$ was attributable to Monte Carlo error from too few resamples rather than a fundamental limitation of the method. The delta method reaches a coverage of 94.6%, falling just short of the nominal target but remaining close.

In terms of precision, the delta method produces narrower intervals on average (12.34 vs. 13.18). This is expected: the delta method assumes a perfectly symmetric normal approximation based on the full sample

size, which yields tighter intervals in large samples. The bootstrap, while more flexible in capturing the true shape of the sampling distribution, introduces additional variability from the resampling process itself, resulting in slightly wider intervals.

Taken together, these results suggest a meaningful tradeoff between the two methods. For large-scale observational datasets such as this one ($n = 10,000$) the delta method offers superior precision with only a marginal sacrifice in coverage. The bootstrap, on the other hand, provides exact nominal coverage but at greater computational cost and with wider intervals. For settings where distributional assumptions are uncertain or sample sizes are smaller, bootstrapping may be the more reliable choice.

Method	Coverage	Mean CI Width
Bootstrap Method (B = 2000)	95.0%	13.18
Delta Method	94.6%	12.34

Figure 2: Coverage and Precision of Delta Method and Bootstrap Confidence Intervals